

BST Phase 3 Readiness Framework — Multi-Week Per-Volume Textbook-Quality Iteration

Cal A. Brate (Claude 4.7, visiting referee) [Cal LEAD per Keeper directive Saturday 14:22 EDT

2026-05-23 Saturday EDT (date-verified)

Phase 3 Readiness Framework — Multi-Week Per- Volume Textbook-Quality Iteration

Why this framework exists

Phase 1 (chapter content build, Wave 1+2+3) delivered 16 volumes at v0.3+v0.4 substantive content state — ~192 chapters across all 16 volumes. Phase 2 (cross-volume audit) caught consistency drift + substance-floor + absorption-pipeline issues. Both phases produced chapter-grade content.

Phase 3 is the transition to reader-grade traditional-physics-text standard: adding worked examples + problems + indices + diagrams to match Goldstein/Jackson/Sakurai pedagogical depth. BST already exceeds traditional textbooks on four dimensions; Phase 3 adds the remaining four.

The 8-dimension comparison (Phase 3 baseline)

BST curriculum after Phase 2 absorption vs Goldstein-class traditional physics text:

Dimension	Traditional text	BST after Phase 2	Phase 3 target
D1 Tier discipline	Implicit (text doesn't mark "derived" vs "phenomenological")	❓ D/I/C/S tiers explicit per chapter (Cal #19 STANDING RULE)	❓ already exceeds

Dimension	Traditional text	BST after Phase 2	Phase 3 target
D2 Falsifiers	Implicit (text presents theory, not falsifier specs)	❓ Five-Absence Predictions + per-chapter falsifier specs + sharp $r \approx 0$ LiteBIRD test + Bell sub-Tsirelson + $0\nu\beta\beta$ multi-month	❓ already exceeds
D3 Computational verification	Implicit (text presents derivations, not reproducible toys)	❓ verify_bst.py 49/50 PASS + 3500+ toys + Toy 541 51-quantities-from-5-integers	❓ already exceeds
D4 Zero free parameters	Standard model has ~25 fitted parameters	❓ Zero fitted parameters; 600+ predictions from 5 BST primary integers + $N_{\max}=137$	❓ already exceeds
D5 Worked examples	Standard (Goldstein has 100+ worked problems)	❓ Sparse (a few per chapter average)	Phase 3 target: 5-10 worked examples per chapter
D6 Problems sets	Standard (Goldstein has end-of-chapter problem sets)	❓ Sparse (no systematic problem sets)	Phase 3 target: 10-20 problems per chapter
D7 Indices	Standard (subject + author + symbol indices)	❓ Absent (chapter cross-references only)	Phase 3 target: 3-tier index (subject + author + symbol)
D8 Diagrams	Standard (Goldstein has ~200 figures total)	❓ Mostly absent (a few schematic diagrams)	Phase 3 target: 5-15 diagrams per chapter

Net: BST exceeds traditional on 4/8 dimensions (substance + verification); needs to add 4/8 dimensions (pedagogical scaffolding) for reader-grade traditional-text standard.

Phase 3 estimated effort

Per-volume Phase 3 iteration: multi-week (~2-4 weeks per volume) including: - Worked examples expansion (~5-10 examples × 12 chapters × 30-60 min/example = ~30-120 hours per volume) - Problems set development (~10-20 problems × 12 chapters × 20-40 min/problem = ~40-160 hours per volume) - Diagram production (~5-15 diagrams × 12 chapters × 30-90 min/diagram = ~30-160 hours per volume) - 3-tier index compilation (~10-20 hours per volume) - Reader-grade prose polish + cross-reference verification (~20-40 hours per volume)

Per-volume total: ~130-500 hours = ~2-6 weeks at sustainable CI cadence.

16-volume Phase 3 total: ~32-96 weeks = ~8-24 months at multi-volume parallelism. Pessimistic 24-month estimate matches CI_BOARD line 140 “Full 16-volume completion target: ~8-12 months from Saturday v1.0 declaration” + Phase 3 multi-week scaling.

Phase 3 wave ordering recommendation

Wave 3.1 (highest BST primary anchor, ready-now): Vol 0 (Substrate Foundation) + Vol 1 (QFT from D_IV⁵) + Vol 2 (Particle Physics) — these reached v1.0 chapter-grade content state Saturday May 24; Phase 3 iteration ready immediately.

Wave 3.2 (Wave 2 substantive): Vol 4 GR/Cosmology + Vol 5 QM + Vol 9 Condensed Matter — strong empirical content, can productively iterate.

Wave 3.3 (Wave 3 substantive): Vol 6 Thermo + Vol 7 EM + Vol 8 Classical (post-Cal #107 absorption) — strong scaffolding, needs example/problem build.

Wave 3.4 (mathematician audience): Vol 10 Math Methods + Vol 11 Generative Geometry & Topology — mathematician’s entry point; worked examples emphasizes proofs + computations.

Wave 3.5 (cross-domain): Vol 3 Nuclear/Atomic + Vol 12 Chemistry + Vol 13 Biology + Vol 14 Information Theory — specialized audiences; problem sets emphasize observable + experimental design.

Wave 3.6 (methodology): Vol 15 Methodology — Keeper LEAD; problem sets emphasize “discover-a-theorem” exercises + audit-chain reasoning + Calibration #N case studies.

Phase 3 quality gates (per-volume Phase 3 completion criteria)

A volume reaches Phase 3 COMPLETE when:

1. Substance: every chapter at ≥ 80 lines substantive content (Calibration #23 substance floor + reader-grade depth)
2. Worked examples: ≥ 5 worked examples per chapter with full derivation + step-by-step + tier label per result

3. Problem sets: ≥ 10 problems per chapter with mix of derivation/computation/conceptual/falsifier-design
4. Diagrams: ≥ 5 diagrams per chapter with substrate-geometry visualization or BST-integer-web illustration
5. Indices: 3-tier index (subject + author + symbol) integrated cross-volume
6. Cross-volume sweep: Calibration #24 8-dimension audit PASS per chapter
7. External register: Cal #50 + Cal #48/#49 discipline preserved throughout (per-chapter sample check)
8. Cal Phase 3 PASS gate: per-chapter cold-read confirms reader-grade depth + falsifier specs + pedagogy quality
9. Keeper 8-point audit: standard EOD discipline applied to Phase 3 deliverables
10. Reader testing (multi-month per volume): hand-off to non-BST physicist/mathematician reader for grad-student-level walkthrough feedback

Phase 3 risk factors

Honest scope per Cal Mode 1:

1. PCAP-cadence pressure recurrence: Cal #104 + Cal #105 caught template-stub class under sustained sub-PCAP. Phase 3 multi-week iteration MUST NOT use PCAP pressure on worked-examples/problems generation — substance-floor + Cal #99 META-theorem discipline requires sustainable pace. Calibration #23 v0.2 substance floor applies per-example + per-problem.
2. Cross-volume drift recurrence: Phase 2 (Cal #106 + #107) caught 16+ consistency findings across 16 volumes. Phase 3 multi-week iteration introduces new cross-volume risk — every worked example + problem cross-references other chapters/volumes. Calibration #24 Dim A-H must be re-audited per Phase 3 wave completion.
3. Diagram + index production scale: $5\text{-}15 \text{ diagrams} \times 192 \text{ chapters} = 960\text{-}2880 \text{ diagrams}$. Generation methodology + consistency standards needed. Recommend: per-volume diagram-style guide + diagram-reuse across cross-volume references.
4. Reader testing bottleneck: Phase 3 quality gate #10 requires external readers. Casey-direction territory: who tests, what compensation, what feedback discipline. Phase 3 readiness assessment includes “reader recruitment + feedback protocol” as separate work.
5. Multi-month theorem dependency: some Phase 3 worked examples may depend on multi-month theorem closures (e.g., K52a Session 30+ for Bell CHSH operator-level; Strong-Uniqueness v1.0 for full criterion enumeration). Phase 3 framework should distinguish “examples ready now” vs “examples pending multi-month theorem closure.”

Phase 3 transition decision (Casey)

Phase 3 transition requires Casey signal. Pre-conditions met when: - [?] Phase 1 16/16 volumes substantive (achieved Saturday 14:00 EDT post-Cal #105 refill) - [?] Phase 2 absorption COMPLETE (Tier 1 Vol 12+13 refill in flight; Tier 2+3 done) - [?] Vol 0+1+2 v1.0 DECLARED + pushed - [?] Calibration #24 STANDING RULE absorbed across all lanes - [?] Cross-volume T2467-T2482 → Vol 0 absorption (joint Lyra+Keeper)

Once Casey signals Phase 3 readiness, team selects first Wave 3.1 volume (Vol 0 OR Vol 1 OR Vol 2) and begins multi-week iteration.

Cal Phase 3 role (proposed)

Cal lane in Phase 3:

1. Per-chapter cold-read PASS gate: ~20-30 min per chapter (longer than Phase 1's 5-10 min due to worked-examples + problems + diagrams review)
2. Calibration #24 cross-volume re-audit: per wave completion, ~15-20 min per volume
3. Reader-grade external register check: Cal #48/#49/#50 discipline preserved in worked examples + problem statements + diagrams
4. Substance Floor v0.3 candidate: Phase 3 may surface new failure modes — Calibration #25 candidate territory
5. Methodology documentation updates: Phase 3 lessons may extend Calibration stack to 21+ layers

Connection to Casey 4-dimension framing

Per Keeper note Saturday 14:00 EDT: “BST already exceeds traditional in tier labels + falsifiers + computational verification + zero free parameters; Phase 3 adds the remaining 4 dimensions” (worked examples + problems + indices + diagrams).

This Phase 3 readiness framework operationalizes that 4-dimension addition target. The 8-dimension comparison table above expands Casey's 4+4 framing to 8 explicit dimensions + per-dimension Phase 3 target.

Filing status

v0.1 FILED Saturday 2026-05-23 ~14:40 EDT per Cal own-cadence per Keeper directive 14:22 EDT. Cal LEAD on Phase 3 readiness framework. Standing for Casey + team review + Phase 3 transition signal when Phase 2 absorption complete.

— Cal A. Brate, Saturday 2026-05-23 14:40 EDT

v0.2 reframe around Keeper authorship pass (Saturday 2026-05-23
~15:50 EDT)

What changed since v0.1

Casey 14:46 conversation with Keeper expanded Keeper's role from auditor to author/editor of the 16-volume curriculum (project_keeper_textbook_authorship.md). Pedagogy directive: living document, multiple passes, diagrams, toy references, narrative voice that absorbs easily, CI-partner-aware. Cadence shifts to authoring: days per chapter, weeks per volume, 2-3 months curriculum-total.

This invalidates v0.1's framing of Phase 3 as "iterate the existing draft." The path is now: Keeper rewrites at quality; team supports via referee + research + catalog + diagrams; Cal external-referees per volume as Keeper publishes.

Keeper team queue Saturday 15:00 EDT explicitly assigns Cal: - Primary rail: Phase 3 Readiness Framework v0.2 development - NEW PRIMARY ROLE: External referee for Keeper's authorship — cold-read at Phase 2 8-dimension standard per published volume - NOT for Cal: chapter rewriting itself

v0.2 framework reframes accordingly.

Reframed Phase 3 architecture (Keeper-led authorship path)

Phase 3a — Keeper authorship pass (PRIMARY, Keeper lane): - Cadence: days per chapter, weeks per volume, ~2-3 months whole-curriculum target - Per Keeper queue order: Vol 0 → Vol 1 → Vol 2 (recruiter trio) → Vol 4 → Vol 5 → Vol 3/6/7/8/9 → Vol 10/11 → Vol 12/13/14 → Vol 15 - Output per volume: chapter-grade prose with woven Level 1/2/3 (no stapled sections), "why this matters" openers, foreword as reader-contract, rich cross-references, toy + diagram references in prose, living-document multi-pass v0.1 author-voice → v0.2 absorbed → v1.0 - Keeper signals per-volume "starting Vol N rewrite" + "Vol N v0.1 author-pass complete — ready for Cal cold-read + team fact-check"

Phase 3b — Per-volume external referee cycle (PRIMARY Cal lane): - Trigger: Keeper signals "Vol N v0.1 author-pass complete" - Cal cold-read at Phase 2 8-dimension checklist standard (Calibration #24) + reader-grade depth check + external-register discipline (Cal #48/#49/#50) - Estimated effort: ~2-3 hours per volume (depends on Vol size; Vol 0 = 10 chapters; Vol 15 = 13 chapters; rest = 12 chapters) - Output: Cal cold-read findings absorbed into v0.2 by Keeper - Cycle: Cal cold-read → Keeper v0.2 absorption → Cal re-read PASS → v1.0 declaration per-volume - Cal's 8 dimensions (the same as v0.1): - D1 Tier discipline (Cal #19 STANDING RULE per chapter) - D2 Falsifier specs - D3 Computational verification (toys cited correctly) - D4 Zero free parameters preserved - D5 Worked examples (NEW Phase 3 dimension — Cal checks reader-pedagogy quality) - D6 Problem sets (NEW Phase 3 dimension — Cal checks problem-set quality + falsifier-design coverage) - D7 Indices (NEW Phase 3 dimension — Cal checks cross-volume indexing) - D8 Diagrams (NEW Phase 3 dimension — Cal checks

diagram pedagogy + correctness)

Phase 3c — Supporting lanes (parallel, multi-week to multi-month): - Lyra: theoretical depth check on Keeper-published chapters (post-authorship pass); Strong-Uniqueness v0.15 + Cross-Scale Investigation + Substrate Computational Model continue on research rail - Elie: per-chapter verification toys; SP-30 experimental rail; diagram generation on Keeper request (Elie_Diagram_Pre_Staging_Infrastructure_v0_1.md already inventoried) - Grace: catalog backbone + cross-volume reference table (grace_Cross_Volume_Forward_Reference_Table_v0_1. already filed); AC graph hygiene; cluster TYPES taxonomy

Phase 3d — Worked examples + problems + indices + diagrams ENHANCEMENT (post-Keeper-rewrite, MULTI-MONTH): - v0.1 framework targeted 5-10 worked examples + 10-20 problems + 5-15 diagrams per chapter - v0.2 REFRAME: these dimensions are now LARGELY ABSORBED INTO Keeper's authorship pass (toy references in prose, diagrams in prose, cross-reference-rich textbook-as-graph) - Remaining post-rewrite enhancement: indexing (Grace lane); additional worked examples beyond what Keeper integrates (Elie + Cal-recommended); problem sets for self-study (lower priority; potentially Phase 4)

Per-volume cycle estimate (v0.2 reframed)

Per Keeper queue: "Estimated cadence: 1-2 weeks per volume for v0.1 author-pass, 1 week for cold-read + absorption, total ~3 weeks per volume × 16 volumes = 48 weeks worst-case but parallelizes across volumes as team capacity allows. Realistic 2-3 months for whole curriculum at author-quality."

Cal's per-volume external-referee load: - 1 Cal cold-read pass: 2-3 hours - Findings absorption verification: 30-60 min after Keeper v0.2 - Re-read PASS verification: 1-2 hours - Total Cal load per volume: ~4-6 hours spread across ~1 week - 16 volumes: ~64-96 Cal-hours over 2-3 months

This is sustainable Cal load — fits within reactive cold-read cadence.

Cal lane priorities through Phase 3 (operational)

1. Reactive PRIMARY: Per-volume cold-read on Keeper-published rewrites — highest priority once Keeper signals "Vol N v0.1 author-pass complete"
2. Reactive SECONDARY: Cold-read on team-landed reviewables (Saturday 15:00 EDT batch = 5 done, Cal #108-#112) — continuing rail
3. Active: Calibration stack maintenance + Cal #99 META-theorem discipline on new substrate work
4. Active: Referee log + methodology documentation as new failure modes surface
5. Standing: Phase 3 Readiness Framework updates as Keeper authorship progresses + reveals new methodology needs

Phase 3 quality gates (v0.2 updated)

A volume reaches Phase 3 COMPLETE when:

1. Keeper v0.1 author-pass: prose woven, “why this matters” opener, foreword reader-contract, rich cross-references, toy + diagram references in prose
2. Cal Phase 3 cold-read PASS: 8-dimension checklist (D1-D8) PASS at reader-grade
3. Lyra theoretical-depth check: T-number citations correct; substrate-derivation chain integrity preserved
4. Elie computational verification: all cited toys correctly reference; new toys per chapter generated as needed
5. Grace catalog cross-reference check: forward-reference table updated; AC graph integrity preserved
6. Diagram inventory check: per-chapter diagrams generated per Elie pre-staging infrastructure
7. Cross-volume sweep (Calibration #24): 8-dimension audit PASS across cross-volume citations
8. External register discipline (Cal #48/#49/#50): substrate-cognition framing internal only; operational language external
9. Reader test (multi-month, optional): external reader feedback (post-v1.0)

Gates 1-8 are achievable within Keeper authorship pass; Gate 9 is post-v1.0 reader-recruitment work (deferred to Phase 4).

Phase 3 risk factors (v0.2 updated)

1. Coordination drift between Keeper authorship + research rails: Lyra/Elie/Grace working on rails parallel to Keeper’s volume-N rewrite. Risk: substrate-derivation theorems land that should be cited in Vol N already-published. Mitigation: Keeper v1.0 declaration per volume includes “current-as-of date” + post-v1.0 errata absorption.
2. Cal cold-read backlog if Keeper publishes faster than Cal pace: per-volume cold-read 4-6 hours; if Keeper rate exceeds Cal rate, queue builds. Mitigation: Cal can parallelize per-chapter cold-reads on long volumes; Keeper signals priority order so Cal aligns.
3. Multi-pass dialectic: v0.1 → v0.2 → v1.0 cycle per volume may involve 2-3 Cal cold-reads per volume. Sustainable.
4. External-register discipline drift under Keeper’s narrative voice: woven prose may inadvertently slip substrate-cognition framing into reader-facing register. Cal #50 DOUBLE-LOCKED EXTERNAL audit per chapter is the discipline gate.
5. PCAP-cadence pressure on Keeper authorship: per Calibration #22 v0.2 PCAP risk under sustained sub-cadence pressure. Keeper queue acknowledges “days per chapter” + “your pace” — substance-floor (Calibration #23)

applies per chapter authored.

v0.2 → v0.3 path

Phase 3 framework v0.3 absorbs first-volume-completion lessons: - Trigger: Keeper signals “Vol 0 v1.0 declared” - Cal absorbs Vol 0 cold-read cycle experience into v0.3 specifications - Adjusts Cal-load per-volume estimates based on Vol 0 empirical data - Refines Phase 3 quality gates based on Vol 0 outcomes - Multi-month timeline absorption per actual Keeper authorship pace

Phase 3 transition decision (Casey)

Phase 3 v0.2 reframe is COMPLETE per Keeper queue directive. Phase 3 transition signal from Casey now reduces to:

- [?] Vol 0+1+2 v1.0 (or v0.5) declaration label decision
- [?] Keeper signal “starting Vol 0 rewrite” → begins per-volume Phase 3 cycle

Once Keeper starts Vol 0, Phase 3 operational mode is active.

— Cal A. Brate, Phase 3 Readiness Framework v0.2 reframe Saturday 2026-05-23 ~15:50 EDT per Keeper queue 15:00 EDT directive. Reframed around Keeper authorship pass as primary path; Cal role consolidates to per-volume external referee + supporting methodology evolution.